

DATA ANALYSIS PORTFOLIO PROJECT

PART 1

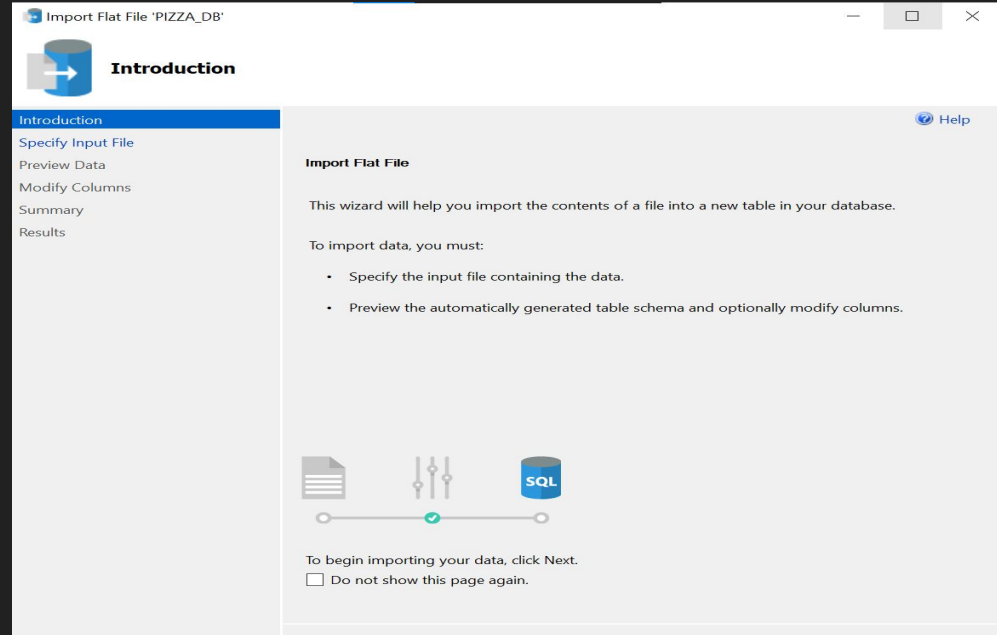
MS SQL SERVER





MS SQL SERVER

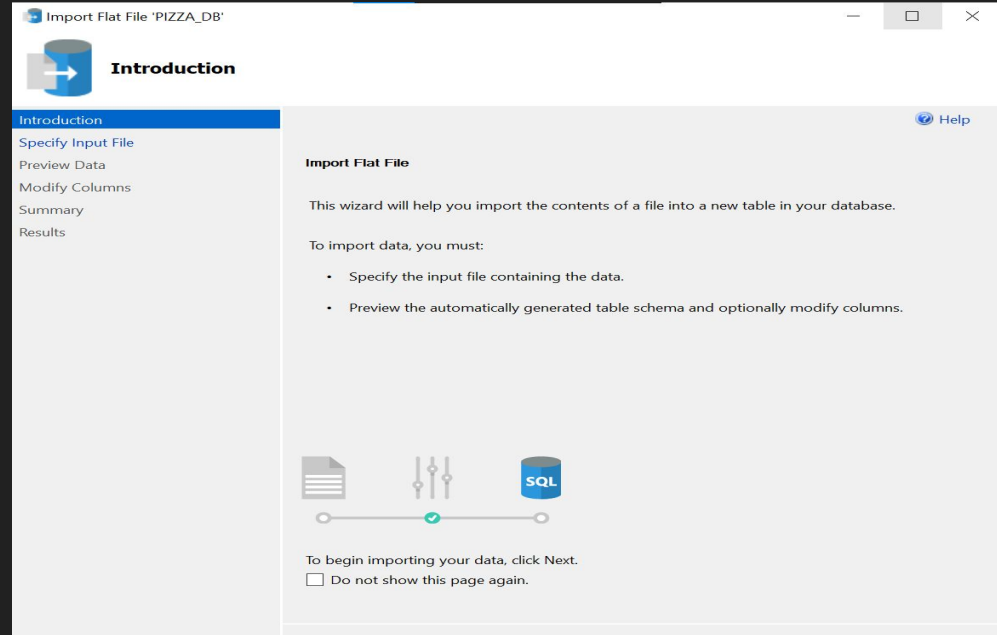
IMPORT DATA





MS SQL SERVER

CREATING DB





MS SQL SERVER

WRITING QUERIES

```
SELECT DATENAME(DW, order_date) AS "Order Day", COUNT(DISTINCT order_id) as "Total Orders"
FROM pizza_sales
WHERE DATEPART(QUARTER, order_date) = 1
GROUP BY DATENAME(DW, order_date)

SELECT DATEPART(HOUR, order_time) as "Order Hours", COUNT(DISTINCT order_id) as "Total Orders"
FROM pizza_sales
GROUP BY DATEPART(HOUR, order_time)
ORDER BY DATEPART(HOUR, order_time)

SELECT pizza_category, CAST(SUM(total_price) AS DECIMAL(10,2)) as "Total Revenue",
CAST(SUM(total_price) *100 / (SELECT SUM(total_price) from pizza_sales) AS DECIMAL(10,2)) AS PCT
FROM pizza_sales
GROUP BY pizza_category
```



MS SQL SERVER

CREATING REPORT



Pizza Sales SQL Queries

PROBLEM STATEMENT

A. KPI's REQUIREMENT

1. Total Revenue :

```
select SUM(total_price) as "Total Revenue"  
from pizza_sales;
```

Results		Messages	
Total Revenue			
1	817860.05083847		

2. Average Order Value :

```
select SUM(total_price)/COUNT(distinct order_id) as "Avg Order Value"  
from pizza_sales;
```

Results		Messages	
Avg Order Value			
1	38.3072623343546		

DATA ANALYSIS PORTFOLIO PROJECT

PART 2

POWER BI

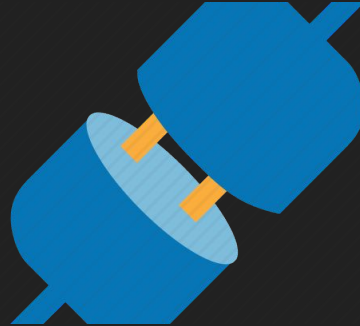




POWER BI

CONNECTING TO

MS SQL SERVER





POWER BI

DATA CLEANING



Pizza Report Dashboard

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Refresh Preview Properties Advanced Editor Choose Remove Keep Remove Split Group Data Type: Whole Number Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Text Analytics Vision Azure Machine Learning AI Insights

Queries [1]

Table: RenameColumns(*"Inserted Month",{"Month", "Month Number"})

	l_name_id	quantity	order_date	order_time	unit_price	total
1	l_m	1	1/1/2015	11:38:36 AM	13.25	
2	x_m	1	1/1/2015	11:57:40 AM	16	
3	se_j	1	1/1/2015	11:57:40 AM	18.5	
4	j	1	1/1/2015	11:57:40 AM	20.75	
5	l_m	1	1/1/2015	11:57:40 AM	16	
6	l	1	1/1/2015	11:57:40 AM	20.75	
7	m	1	1/1/2015	12:12:28 PM	16.5	
8	x_l	1	1/1/2015	12:12:28 PM	20.75	
9	m	1	1/1/2015	12:16:31 PM	16.5	
10	m	1	1/1/2015	12:21:30 PM	16.5	
11	s	1	1/1/2015	12:29:36 PM	12.75	
12	s	1	1/1/2015	12:29:36 PM	12	
13	upr_s	1	1/1/2015	12:50:37 PM	12.5	
14	upr_s	1	1/1/2015	12:51:37 PM	12.5	
15	s_s	1	1/1/2015	12:52:01 PM	12	
16	den_s	1	1/1/2015	12:52:01 PM	12	
17	j_l	1	1/1/2015	12:52:01 PM	20.5	
18	j	1	1/1/2015	12:52:01 PM	20.75	
19	s	1	1/1/2015	12:52:01 PM	12.5	
20	s	1	1/1/2015	12:52:01 PM	12	
21	l	1	1/1/2015	12:52:01 PM	20.75	

16 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON SATURDAY

Query Settings

PROPERTIES

Name

pizza_sales

All Properties

APPLIED STEPS

Source

Navigation

Replaced Value

Replaced Value1

Replaced Value2

Replaced Value3

Filtered Rows

Inserted Day Name

Added Conditional Column

Renamed Columns

Inserted Month Name

Inserted Month

Renamed Columns1

ENG 5:13 PM 5/6/2025



POWER BI

DATA PROCESSING

[illegible]



POWER BI

DATA

VISUALIZATION



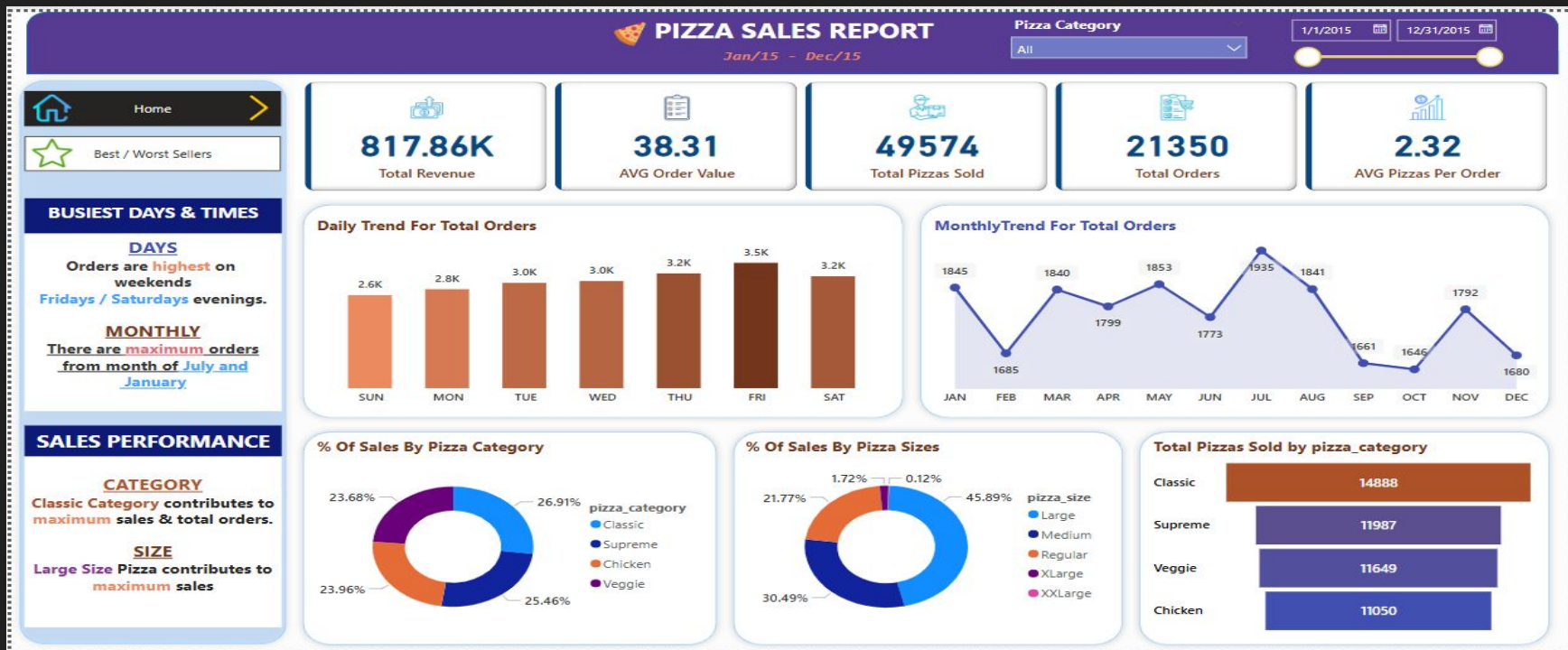
MonthlyTrend For Total Orders





POWER BI

REPORT / DASHBOARD





POWER BI

REPORT / DASHBOARD



PROBLEM STATEMENT

KPI's REQUIREMENT

We need to analyze key indicators for our pizza sales data to gain insights into our business performance. Specifically,

We want to calculate the following metrics :

1. **Total Revenue** : The sum of total price of all pizza orders.
2. **Average Order Value** : The average amount spent per order, calculated by dividing the total revenue by the total no of orders.
3. **Total Pizzas sold** : The sum of quantities of all pizzas sold.
4. **Total Orders** : The total no of orders placed.
5. **Average Pizzas per order** : The average no of pizzas sold per order, calculated by dividing the total no of Pizzas sold by the total no of orders

PROBLEM STATEMENT

CHARTS REQUIREMENT

We would like to visualize various aspects of our pizza sales data to gain insights and understand key trends.

We have identified the following requirements for creating the charts :

1. Daily Trend For Total Orders :

Create a bar chart that displays the daily trend of total orders over a specific time period. This chart will help us identify any patterns or fluctuations in order volumes on a daily basis

2. Monthly Trend For Total Orders :

Create a line chart that illustrates the hourly trend of total orders throughout the day. This chart will allow us to identify peak hours or periods of high order activity.

3. Percentage of Sales by Pizza Category :

Create a pie chart that shows the distribution of sales across different pizza

Categories. This chart will provide insights into the popularity of various pizza categories and their contribution to overall sales.

PROBLEM STATEMENT

CHARTS REQUIREMENT

4. Percentage of Sales by Pizza Size :

Generate a pie chart that represents the percentage of sales attributed to different pizza sizes. This chart will help us understand customer preferences for pizza sizes and their impact on sales.

5. Total Pizzas sold by Pizza Category :

Create a funnel chart that presents the total no of pizzas sold for each pizza category. This chart will allow us to compare the sales performance of different pizza categories.

6. Top 5 Best Sellers By Revenue, Quantity and Total Orders :

Create a bar chart highlighting the top 5 best selling pizzas based on the Revenue, Total Quantity, Total Orders. This chart will help us understand the most popular pizza options.

PROBLEM STATEMENT

CHARTS REQUIREMENT

7. Bottom 5 Worst Sellers By Revenue, Quantity and Total Orders :

Create a bar chart showcasing the bottom 5 worst selling pizzas based on the Revenue, Total Quantity, Total Orders. This

Chart will enable us to identify the underperforming or less popular pizza options.

DATA ANALYSIS PORTFOLIO PROJECT

SOFTWARE USED :

MS OFFICE / EXCEL : Version 2408 Build 16.0.17932.20328

MS SQL SERVER 2022 : 16.0.4185.3

MS SQL SERVER MANAGEMENT STUDIO : 20.2.1

POWER BI : Version: 2.139.1733.0 January 2025